Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			**Indicative content:** Both cans the same distance [5 cm] from the heater *If one can is closer to the heater it will be heated more than the other* Same volume of water [200 cm^3] in both cans *If different volumes used, the temperature rise would be different* Same starting temperature [8 ^oC] of water *Different starting temperatures of water will affect final temperature* Same heater used to heat both cans *Both cans heated by the same power output* Both cans have lids *Reduces convection from both cans* Both cans contain water *If different liquids used, the temperature rise would be different* Both cans identical [size / material], apart from colour *Different sizes / materials may absorb radiation differently* Both cans heated for the same time [8 min] *Energy from the heater same for both cans* **5–6 marks** Identify 3 controlled variables **with** correct explanations. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.*	3	3		6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				3–4 marks At least 3 controlled variables **or** correct explanations. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1–2 marks** Identified up to 2 controlled variables **or** correct explanations. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*. *The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** – *No attempt made or no response worthy of credit.*						
	(b)	(i)		Correctly plotted point at (5, 25) < small square tolerance (1) Best fit curve from 1 – 8 minutes < small square tolerance (1) Don't accept thick, wispy, disjointed, double lines		2		2		2
		(ii)		The black can has a {bigger / faster / higher} temperature [rise] (1) Black surfaces are {better absorbers / poor reflectors} [of heat radiation] (1) N.B. converse arguments accepted			2	2		2
				Question 5 total	**3**	**5**	**2**	**10**	**0**	**10**